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Beyond the Hype: Navigating the Conservation Implications of Artificial Intelligence

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ABSTRACT

Conservation AI—the deliberate application of artificial intelligence technology to achieve conservation goals—has great potential to boost productivity, make existing conservation actions more efficient, and enable entirely new areas of activity. However, it also comes with risks, including AI being used by bad actors; high material demand for energy, land, and water; biases in training datasets; AI-fueled techno-optimism distracting from other actions; and undesirable changes in staffing and working practices in the conservation sector. Changes in wider society brought about by AI in areas such as agriculture, human health, and labor markets may also have significant impacts on biodiversity (whether positive or negative), as these are major drivers of biodiversity loss. This article reviews the various links between AI and conservation, arguing that to date there has been too much techno-optimism and a lack of attention to risks and broader implications. It concludes with recommendations for how conservation could approach AI more effectively by considering risks and potential unintended consequences; adopting a principle of transparency; ensuring AI does not harm the staff, skills, and independence of the conservation sector; and investing in research and advocacy to address the conservation implications of wider societal changes caused by AI.

1 | Introduction

Hype around new technology is a familiar part of modern life. However, nothing compares to the interest in artificial intelligence (AI) following the launch of OpenAI's ChatGPT in November 2022. This triggered an avalanche of applications and news stories, ranging from concern about students cheating on exams to doomsday predictions of robots taking over the world. While some forms of AI have been in use for decades, human society is still in the early stages of adjusting to generative AI (AI that is capable of generating new content, such as images or text, in response to a submitted prompt; Merriam-Webster 2024). From the mundane (writing emails and planning meetings) to the systemic (reshaping labor markets and demographic trends), it seems that there are few aspects of human life that will not be touched by AI.

But what of nonhuman life? The conservation sector has been quick to leap onto the AI bandwagon, with multiple new tools, AI-focused organizations, and funding mechanisms that have been proposed or established. These are part of what can be thought of as Conservation AI—the deliberate application of AI technology to achieve conservation goals. While Conservation AI is undoubtedly exciting, there are several potential risks and unintended consequences that should be taken seriously. There are also broader implications of AI for life on earth, including the downstream effects on biodiversity of AI-induced societal transformations. In this article, I consider these different links between AI and conservation, arguing that to date there has been a lack of attention to risks and broader implications. I conclude with recommendations for how conservation could approach AI more critically, recognizing its risks and weaknesses as well as its strengths (Figure 1).

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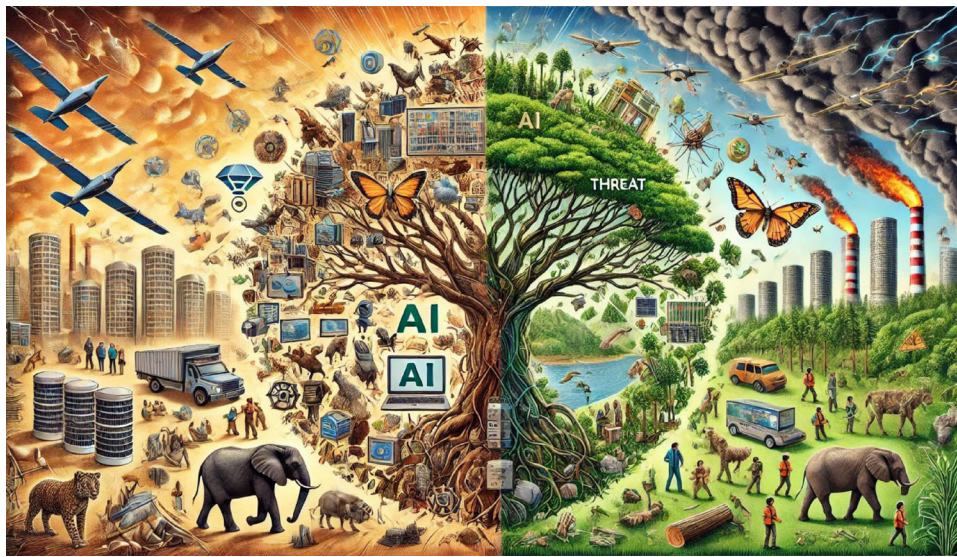


FIGURE 1 | A visual summary of the full text of this article, generated by ChatGPT 4o.

2 | Applications of Conservation AI

There are three broad ways in which AI might be deployed for conservation. First, the conservation sector will benefit from generative AI applications that increase general productivity. Examples include tools to generate text, summarize documents, manage email inboxes, write better code, manage files, communicate across languages, schedule meetings, and enhance institutional memory (e.g., Chandra 2024). AI tools for creating images and videos, such as *MidJourney*, could also be used to create persuasive marketing and communication content for effective behavior change and fundraising campaigns.

Second, bespoke Conservation AI tools can be applied to existing conservation activities to make them faster, cheaper, and more efficient. To date, this is mostly being done with machine learning algorithms rather than newer generative AI models. Examples include extracting information about wildlife from camera trap images (Vélez et al. 2023), modeling the expansion of road networks from satellite images (Sloan et al. 2024), and synthesizing conservation evidence (Berger-Tal et al. 2024). Many of these help to address the challenge of processing vast volumes of data that are now routinely collected by monitoring and surveillance devices (Kwok 2019). Conservation AI applications of this kind are generating considerable media interest (e.g., Green 2022) and driving the emergence of a whole new sector of conservation tech startups such as *Conservation AI* and *Wildlife AI* (e.g., see Dehmler-Buckley and Palermo 2024).

Third, a potentially more transformative approach to Conservation AI is using it to do entirely new things. This aligns with the view of Sam Altman, CEO of OpenAI, who has said, “I totally believe that if you give people way more powerful tools, it’s not just that they can work a little faster, they can do qualitatively different things.” (Unconfuse me 2024). For example, generative AI could be used to radically improve communication with humans who are excluded from conservation decisions due to language barriers. It could also be used for interspecies communication, either directly by “translating”

nonhuman communication (e.g., the *Earth Species Project*) or by simulating nonhuman perspectives using trained large language models (e.g., *Nature Perspectives* enabling visitors to talk to zoology museum exhibits; Sample 2024). These tools could be used to increase understanding and empathy for nonhuman life, including in governance processes such as planning decisions where nonhumans might be given a (simulated) seat at the table (Bakker 2022).

3 | Risks of Conservation AI

While the opportunities set out above are exciting, there are multiple ways in which Conservation AI could generate unintended and/or undesirable consequences.

First, AI opens a new frontier in an ongoing technological arms race between conservation authorities and their opponents. Conservation AI systems that predict the movements of wildlife or illegal hunters can be adapted to become tools for improving hunting efficiency or evading law enforcement authorities. These anticonservation applications can, again, be divided into those that “improve” on existing approaches and those that generate entirely new problems for conservation. Examples of the latter might include the creation of deepfake content apparently showing trusted individuals or organizations disseminating anti-conservation messages (as has been demonstrated for climate change; Doss et al. 2023) or large corporations using AI-generated content to greenwash their image or lobby against regulation. While very little is currently known about how actors hostile to conservation might use AI, it would be naïve of the conservation sector to overlook this potential threat.

Second, despite the benign language of terms like “cloud computing” (Nost and Colven 2022), AI systems have a high demand for energy, water, land, and other resources that could directly undermine conservation goals. The computational power required by AI doubles roughly every 100 days (Zhu et al. 2023), and the energy needs are substantial—particularly for generative AI.

Most current Conservation AI is based on machine learning with lower material needs, but future applications are likely to make more use of generative tools.

Third, it is well established that the datasets used to train AI systems contain significant biases, and these could be consequential for conservation (Wearn, Freeman, and Jacoby 2019). This is true of both ecological (Ellis-Soto, Chapman, and Locke 2023) and large language model datasets. For example, in a recent study asking ChatGPT questions about ecological restoration and stakeholder engagement, Urzedo et al. (2024) found that “more than two-thirds of the chatbot’s answers rely on the expertise of male academics working at universities in the United States, while largely ignoring evidence from low- and lower-middle-income countries (7%) and Indigenous and community restoration experiences (2%).” This is particularly concerning given that it contradicts recent efforts to promote the rights and sovereignty of Indigenous peoples and local communities in conservation policy and practice, and to recognize the importance of plural and diverse knowledge in conservation (McElwee et al. 2020). The emerging field of conservation data justice is doing important work to promote the ethical collection, storage, and use of conservation data (e.g., Pritchard et al. 2022).

Fourth, there is a risk that AI will be used to generate advice or summaries of evidence that do not take account of the specific local nature of conservation problems or the full range of relevant information. Despite appearances, AI does not provide objective or value-free technical solutions to conservation problems, which are often inherently political, value-laden, and deeply place-based (Woodrich and Pape 2024). The “black box” nature of AI systems, which most users (and even in some cases their designers) do not fully understand, might contribute to this problem. A related concern is that techno-optimism generated by the hype around AI could provide a false sense of security, lulling citizens and policymakers into the view that there is no need to worry about pressing conservation challenges because AI is going to solve them all, or pulling resources away from important but less novel activities.

Fifth, it can be hard to predict the full range of impacts of new technologies. For example, social media was initially seen as empowering and democratizing but has become a problematic form of surveillance capitalism. This is particularly challenging with AI, because many potential applications are so new. To give a speculative example, AI-generated virtual ecotourism experiences intended to connect people to nature could instead create a dissociation with “real” nature because the virtual alternative is preferable. Indeed, newly emerging digital natures could become so convincing and ubiquitous that they challenge the very idea of what we mean by “nature” itself (Luque-Ayala, Machen, and Nost 2024). More prosaically, AI systems can generate weird or embarrassing outputs, particularly when they are not checked first by a human reviewer.

Finally, Conservation AI might result in changes in the working practices, structure, and function of the conservation sector. Automation is likely to affect certain types of jobs or staff members more than others and may result in further centralization of decision-making in data centers and HQ buildings far from the field, undermining hands-on, place-based experience

of conservation practice (Adams 2019) as well as efforts to make conservation more equitable, diverse, and inclusive. The emergence of new Conservation AI companies (such as *NatureMetrics* and *Biometrio.earth*) also suggests a possible shift in the political economy of conservation toward the private sector.

4 | Broader Conservation Implications of AI

The major drivers of biodiversity loss are ultimately linked to the ways in which human societies and economies interact with nonhuman life on the earth. Therefore, if AI triggers significant changes to our societies and economies, these will have important implications for biodiversity conservation—whether for good or ill. While it is impossible to predict how these processes will unfold, several broad areas can be anticipated.

First, as described above, AI systems themselves have significant material impacts that are likely to negatively impact biodiversity. While AI systems might help to make energy use in the wider economy more efficient (Olatunde et al. 2024), their direct material impacts are likely to outweigh these gains.

Second, AI is likely to have a substantial impact on agriculture, the single largest driver of biodiversity loss. AI has great potential to be used as part of smart, precision agriculture systems that could greatly increase yields (e.g., Zhang and Kovacs 2012) and could make agricultural commodity chains more efficient, reducing wastage and demand for land. These could, in theory, be used to “spare” land for conservation, although this requires a range of targeted policy mechanisms and the political will to match (Phalan et al. 2016). AI and related “smart” agriculture tools may also reinforce existing power hierarchies and undermine local knowledge, particularly in the case of smallholder farmers (Foster et al. 2023).

Third, AI is expected to have a big impact on labor markets, as AI systems become equivalent or superior to humans on a growing range of tasks. While some new jobs for humans will emerge as old ones disappear, it is predicted that AI-driven shifts in labor markets will happen much faster and at a greater scale than in previous technological revolutions (Cazzaniga et al. 2024). This could result in high levels of unemployment, including of previously well-paid workers in white-collar jobs, creating problems for governments and the industries that rely on consumers with disposable income to buy their products. One possible consequence could be the widespread introduction of Universal Basic Income (UBI). What these changes might mean for conservation (and capitalism; Verdegem 2024) is hard to predict, but they could significantly alter patterns of consumption of goods and services, both important drivers of biodiversity loss.

Fourth, AI could lead to changes in human health and settlement patterns. AI could have profound impacts on medicine, potentially extending human life expectancy. Automation of jobs could also lead to changes in where people choose to live, perhaps driving urbanization as farm jobs are mechanized or out-migration from cities when there is no longer any need to live close to an office. Again, predicting these outcomes is difficult, but large shifts in life expectancy and where people choose to

live would all have impacts on biodiversity through processes of land-use change and resource consumption.

Fifth, AI is likely to result in the further centralization of power and wealth in digital capitalism. It is already dominated by a small number of giant software companies (e.g., OpenAI, Google, Microsoft) and chip manufacturers (e.g., Nvidia), with some experiencing dizzying increases in share value over recent years. The further concentration of power and wealth among a small number of companies could make them, and their environmental impacts, increasingly difficult to regulate (Nost and Colven 2022).

5 | Toward More Productive Engagement With AI

The examples set out above demonstrate that the implications of AI for conservation go far beyond enhancing existing conservation activities. In many ways these broader implications are exciting—they could plausibly result in reduced pressures on biodiversity and entirely novel ways to engage people with nonhuman nature. However, there are significant risks, some of which could be catastrophic for biodiversity.

How is the conservation sector engaging with these issues? To judge from the current literature (peer-reviewed, blogs, news articles, reports, etc.) and my conversations with conservation practitioners, there is a lot of enthusiasm for Conservation AI. However, there seems to be minimal consideration of the risks and unintended consequences that it might generate (see Wearn, Freeman, and Jacoby 2019; WWF 2024 for exceptions) and almost no consideration of the wider conservation implications of AI set out in the previous section. I find this alarming, particularly as some of the wider societal transformations that AI could generate could be much more consequential for conservation (whether positive or negative) than technical applications of Conservation AI.

To address these gaps in current engagement between the conservation sector and AI, I offer four recommendations for policy and practice:

1. Developers and users of Conservation AI should ensure they consider the risks, biases, and potential unintended consequences of AI. There are many ways in which Conservation AI could go wrong. While some of these are difficult to predict, it seems clear that Conservation AI is not going to be a solve-all panacea for conservation. Rather, it is another tool that will, like others, have advantages and disadvantages. The conservation sector should actively engage in efforts to mitigate problems with AI, such as finding ways to incorporate Indigenous knowledge and perspectives into AI systems (Lewis 2020).
2. Conservation organizations and researchers should be transparent in reporting how they are using AI in their work. Many journals already require authors to document their use of AI tools in data collection, analysis, and writing. Applying this principle of transparency to all aspects of conservation would enable those using or affected by it to know what has been done by a human or by a machine. This is important for ethical practice. For example, knowing that evidence

synthesis has been done by an AI tool would enable a better understanding of risks of introduced biases (Urzedo et al. 2024).

3. Conservation organizations should consider how adopting AI systems and funding into their working practices might change the structure and function of the conservation sector. Without careful thought, there is a risk that the adoption of Conservation AI will generate undesirable impacts such as job losses or loss of place-based knowledge for people working in and affected by the conservation sector. If AI companies become major funders of conservation, there is also a risk that the donor power that this will give them might stifle critical scrutiny of Conservation AI.
4. Conservation actors need to invest in understanding and influencing how AI is affecting wider society. This should involve new research programs that go far beyond technical studies of AI systems, including collaboration with social scientists (working on ethics, philosophy, political ecology, existential risk, economics, etc.) to understand the wider implications. Conservation actors should also get more involved in advocacy around the development of regulatory frameworks for AI, which will be crucial in guiding (and constraining) the actions of corporations and other developers of AI (e.g., Verdegem 2024). This will require novel partnerships with civil society organizations outside the conservation sector. Key questions here include: How might AI-induced changes in agriculture, labor markets, and human health affect conservation? How can actors from those sectors be engaged to ensure conservation impacts are considered? How might AI be regulated in a way that maximizes benefits of AI for conservation while minimizing risks?

AI promises transformative changes to human society and the ways in which it affects nonhuman life on earth. The conservation sector is right to embrace AI tools where they can be used to achieve its goals. However, it is essential that this is done with a critical mindset—interrogating risks and opportunities carefully, rather than being swept along on a wave of techno-optimism. In this early stage of AI adoption, there are many more questions than answers, making research vitally important to how the sector responds. Conservation needs to step up to meet these challenges, or it risks being left as a bystander, too worried about spotting animals in a camera trap image to notice that the world has changed.

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Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

References

- Adams, W. M. 2019. "Geographies of Conservation II: Technology, Surveillance and Conservation by Algorithm." *Progress in Human Geography* 43: 337–350.
- Bakker, K. 2022. *The Sounds of Life: How Digital Technology Is Bringing Us Closer to the Worlds of Animals and Plants*. Princeton, NJ: Princeton University Press.
- Berger-Tal, O., B. B. M. Wong, C. A. Adams, et al. 2024. "Leveraging AI to Improve Evidence Synthesis in Conservation." *Trends in Ecology & Evolution* 39: 548–557.
- Cazzaniga, M., F. Jaumotte, L. Li., et al. 2024. "Gen-AI: Artificial Intelligence and the Future of Work [WWW Document]." IMF. <https://www.imf.org/en/Publications/Staff-Discussion-Notes/Issues/2024/01/14/Gen-AI-Artificial-Intelligence-and-the-Future-of-Work-542379>
- Chandra, D. 2024. The 14 Best AI Productivity Tools in 2024. The Process Hacker. <https://theprocesshacker.com/blog/ai-productivity-tools/>
- Dehmler-Buckley, O., and P. Palermo 2024. *Nature Fintech Sector Map*. Nature Tech Collective.
- Doss, C., J. Mondschein, D. Shu, et al. 2023. "Deepfakes and Scientific Knowledge Dissemination." *Scientific Reports* 13: 13429.
- Ellis-Soto, D., M. Chapman, and D. H. Locke. 2023. "Historical Redlining Is Associated With Increasing Geographical Disparities in Bird Biodiversity Sampling in the United States." *Nature Human Behaviour* 7: 1869–1877.
- Foster, L., K. Szilagyi, A. Wairegi, C. Oguamanam, and J. de Beer. 2023. "Smart Farming and Artificial Intelligence in East Africa: Addressing Indigeneity, Plants, and Gender". *Smart Agricultural Technology*, 3: 100132.
- Green, G. 2022. "Five Ways AI is Saving Wildlife—From Counting Chimps to Locating Whales." The Guardian, February 21, 2022.
- Kwok, R. 2019. "AI Empowers Conservation Biology." *Nature* 567: 133–134.
- Lewis, J. E. 2020. "Indigenous Protocol and Artificial Intelligence Position Paper." The Initiative for Indigenous Futures and the Canadian Institute for Advanced Research (CIFAR), Hawai'i: Honolulu.
- Luque-Ayala, A., R. Machen, and E. Nost. 2024. "Digital Natures: New Ontologies, New Politics?" *Digital Geography and Society* 6: 100081.
- McElwee, P., Á. Fernández-Llamazares, Y. Aumeeruddy-Thomas, et al. 2020. "Working With Indigenous and Local Knowledge (ILK) in Large-Scale Ecological Assessments: Reviewing the Experience of the IPBES Global Assessment." *Journal of Applied Ecology* 57: 1666–1676.
- Merriam-Webster. (2024). "Generative AI." [Merriam-Webster.com Dictionary](https://www.merriam-webster.com/dictionary/generative%20ai).
- Nost, E., and E. Colven. 2022. "Earth for AI: A Political Ecology of Data-Driven Climate Initiatives." *Geoforum* 130: 23–34.
- Olatunde, T. M., A. C. Okwandu, D. O. Akande, and Z. Q. Sikhakhane. 2024. "Reviewing the Role of Artificial Intelligence in Energy Efficiency Optimization." *Engineering Science & Technology Journal* 5: 1243–1256.
- Phalan, B., R. E. Green, L. V. Dicks, et al. 2016. "How Can Higher-Yield Farming Help to Spare Nature?" *Science* 351: 450–451.
- Pritchard, R., L. A. Sauls, J. A. Oldekop, W. A. Kiwango, and D. Brockington. 2022. "Data Justice and Biodiversity Conservation." *Conservation Biology* 36: e13919.
- Sample, I. 2024. "AI Gives Voice to Dead Animals in Cambridge Exhibition." The Guardian, October 14, 2024.
- Sloan, S., R. R. Talkhani, T. Huang, J. Engert, and W. F. Laurance. 2024. "Mapping Remote Roads Using Artificial Intelligence and Satellite Imagery." *Remote Sensing* 16: 839.
- Unconfuse me. 2024. Unconfuse me with Bill Gates: Interview with Sam Altman [WWW Document]. <https://www.gatesnotes.com/Unconfuse-Me-podcast-with-guest-Sam-Altman>
- Urzedo, D., Z. T. Sworna, A. J. Hoskins, and C. J. Robinson. 2024. "AI Chatbots Contribute to Global Conservation Injustices." *Humanities and Social Sciences Communications* 11: 1–8.
- Vélez, J., W. McShea, H. Shamon, et al. 2023. "An Evaluation of Platforms for Processing Camera-Trap Data Using Artificial Intelligence." *Methods in Ecology and Evolution* 14: 459–477.
- Verdegem, P. 2024. "Dismantling AI Capitalism: The Commons as an Alternative to the Power Concentration of Big Tech." *AI & Society*, 39: 727–737.
- Wearn, O. R., R. Freeman, and D. M. P. Jacoby. 2019. "Responsible AI for Conservation." *Nature Machine Intelligence* 1: 72–73.
- Woodrich, S. T., and T. Pape. 2024. "Ecological Restoration and Artificial Intelligence: Whose Values Inform a Project?" *Restoration Ecology* 32: e14128.
- WWF. 2024. "Artificial Intelligence and Conservation" <https://www.worldwildlife.org/pages/artificial-intelligence-and-conservation>
- Zhang, C., and J. M. Kovacs. 2012. "The Application of Small Unmanned Aerial Systems for Precision Agriculture: A Review." *Precision Agriculture* 13: 693–712.
- Zhu, S., T. Yu, T. Xu, et al. 2023. "Intelligent Computing: The Latest Advances, Challenges, and Future." *Intelligent Computing* 2: 0006.